THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Prime Time** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The smallest prime number that is a factor of 221 is: 1

(a) 17 **(b)** 1 **(c)** 13 **(d)** 15

2. Which of the following is divisible by 11? 1

(a) 2728 **(b)** 2731 **(c)** 2729 **(d)** 2730

3. Which pair is a pair of co-prime numbers? 1

(a) 24 and 36 **(b)** 14 and 35 **(c)** 21 and 28 **(d)** 21 and 22

4. Which of these is a pair of twin primes? 1

(a) 53 and 59 **(b)** 41 and 43 **(c)** 43 and 47 **(d)** 47 and 53

5. The number of prime numbers between 60 and 100 is: 1

(a) 7 **(b)** 10 **(c)** 8 **(d)** 9

6. The prime factorisation of 210 has how many prime factors (counting repeats)? 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 3

7. In the Sieve of Eratosthenes up to 150, we need to cross out multiples of primes up to: 1

(a) 13 **(b)** 7 **(c)** 11 **(d)** 75

8. The smallest prime number that is a factor of 391 is: 1

(a) 19 **(b)** 23 **(c)** 17 **(d)** 1

9. Which of the following is divisible by 4? 1

(a) 7924 **(b)** 7925 **(c)** 7927 **(d)** 7926

10. Which of these is a pair of twin primes? 1

(a) 61 and 67 **(b)** 59 and 61 **(c)** 67 and 71 **(d)** 79 and 83

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Are 25 and 36 co-prime? Give the reason. 2

12. Test whether 9136 is divisible by 4 and by 8. 2

13. Write two prime numbers whose sum is 56. 2

14. Are 21 and 49 co-prime? Give the reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Use the divisibility test for 11 to check whether 8173 is divisible by 11. 3

16. Draw a factor tree (describe) for 198 and write its prime factorisation. 3

17. Find the first 3 common multiples of 3 and 7. 3

OR

Use the divisibility test for 11 to check whether 9625 is divisible by 11. 3

18. Draw a factor tree (describe) for 240 and write its prime factorisation. 3

OR

Find the first 3 common multiples of 8 and 12. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find all prime numbers between 31 and 70 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

OR

A number is divisible by both 7 and 9. Show that it is divisible by 63 when 7 and 9 are co-prime. Give an example with 126. 5

20. Find all prime numbers between 50 and 90 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

OR

A number is divisible by both 5 and 8. Show that it is divisible by 40 when 5 and 8 are co-prime. Give an example with 240. 5

END OF QUESTION PAPER — Total: 40 Marks

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